Slender Object Scene Segmentation in Remote Sensing Image Based on Learnable Morphological Skeleton With Segment Anything Model

Jun Xie, Wenxiao Li, Faqiang Wang, Liqiang Zhang, *Member, IEEE*, Zhengyang Hou, and Jun Liu

***Abstract*— Morphological methods play a crucial role in remote sensing image processing, due to their ability to capture and preserve small structural details. However, most of the existing deep learning models for semantic segmentation are based on encoder–decoder architectures including U-Net and segment anything model (SAM), where the downsampling process tends to discard fine details. In this article, we propose a new approach that integrates learnable morphological skeleton prior (MorSP) into deep neural networks using the variational method. To address the difficulty in backpropagation in neural networks caused by the nondifferentiability presented in classical morphological operations, we provide a smooth representation of the morphological skeleton and design a variational segmentation model integrating MorSP by employing operator splitting and dual methods. Then, we integrate this model into the network architecture of SAM, which is achieved by adding a token to mask decoder and modifying the final sigmoid layer, ensuring the final segmentation results preserve the skeleton structure as much as possible. The experimental results on remote sensing datasets, including buildings, roads, and water bodies, demonstrate that our method outperforms the original SAM on slender object segmentation and exhibits better generalization capability.**

***Index Terms*— Image segmentation, morphological skeleton, remote sensing image, soft threshold dynamics (STD).**

I. Introduction

SEMANTIC segmentation is a fundamental problem in remote sensing image analysis and other fields, aiming to separate objects of interest from the background at the pixel level. This technique has been widely utilized in various practical applications, including urban planning, damage detection, land cover classification, and resource management.

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At present, the mainstream image segmentation methods can be broadly classified into two categories: model-based methods and data-driven methods. The former typically regards the image segmentation as an optimization problem of energy minimization, such as the Potts model [1] and the Snakes model [2]. These energy functions generally consist of two primary components: a fidelity term that ensures the accuracy of segmentation and a regularization term that preserves essential spatial features within the segmentation result. However, the design of traditional variational models requires substantial expertise in related fields and often suffers from poor generalization. Data-driven methods can overcome these limitations by leveraging deep learning techniques. Some neural networks were proposed in the last few years like fully convolutional network (FCN) [3], U-Net [4], and Vision Transformer (ViT) [5], which significantly improved the flexibility and adaptability of image segmentation.

Unlike natural images, segmentation in remote sensing images requires preserving local slender structures and overall topological properties, such as connectivity, holes, and branches, which makes this task particularly challenging. The morphological skeleton [6] is a simplified representation of the object's structure that can accurately describe these topological features. Several studies have incorporated skeletons as prior knowledge in segmentation models. For instance, Hu and Zu [7] utilized skeletons as the initial values for the Snakes model to guide road extraction. Additionally, Gaetano et al. [8] proposed a segmentation method based on the watershed transform [9], where skeletons were used as markers. Skeleton prior has also been integrated into data-driven models. Shit et al. [10] improved the skeleton extraction algorithm and designed a loss function that combines traditional segmentation loss with skeleton loss. Similarly, Xu et al. [11] proposed a connectivity-preserving loss based on a skeleton for the detection of line-shaped objects. Unlike the above methods, Liu et al. [12] designed a convolutional neural network (CNN) to predict road surface, edges, and center lines, and correlated these results to enhance road extraction. However, all these methods lack reliable mathematical interpretability, and whether the subnetworks or loss functions can learn the desired features depends entirely on empirical evidence.

In recent years, foundational models like GPT [13] and CLIP [14] have transformed the field of natural language pro-

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cessing. Recently, Meta AI Research [Facebook AI Research (FAIR)] introduced a visual foundational model called segment anything model (SAM) [15]. After pretraining on the SA-1B dataset, SAM can segment any type of object using prompt inputs such as points, boxes, masks, or text. It demonstrates strong zero-shot generalization performance across multiple remote sensing datasets [16]. Currently, two main approaches exist for adapting SAM to downstream tasks: the first is to design one-shot or few-shot methods to enhance SAM's performance by learning from a limited number of examples in new datasets [17], [18]. The second approach involves adding adaptors or prompters [19], [20] to SAM's network architecture, tailoring outputs for specific tasks, but this lacks a rigorous mathematical foundation.

Most segmentation networks utilize an encoder–decoder architecture to extract high-dimensional features. However, the upsampling and downsampling processes in this design often lead to the loss of fine details, limiting the network's ability to segment small targets or structures. To address this issue, we integrate morphological skeleton prior (MorSP) into the neural network to enhance segmentation accuracy and preserve overall geometric structure. There are primarily three strategies to achieve this. The first strategy is to postprocess the neural network outputs using traditional models. For instance, Zhu et al. [21] utilized conditional random field (CRF) to extract spatial neighborhood information of buildings, which was fused with the network's output to ensure clear segmentation boundaries. Yet, this approach does not correct inherent errors in the neural network's output. The second strategy focuses on constructing a loss function that incorporates spatial priors to guide the model's learning process. For example, Ma et al. [22] introduced an object consistency loss and a boundary preservation loss to fine-tune SAM. However, these models may struggle when input data significantly deviates from the training data, as they cannot use learned prior knowledge during prediction. The third approach unrolls the variational segmentation algorithm to design network layers, integrating prior constraints through regularization and penalty terms into the model. This technique combines the advantages of both model-based and data-driven methods, enabling the network to automatically minimize the energy function during forward propagation. Monga et al. [23] described this technique in deep learning, and numerous studies have explored its potential [24], [25], [26], [27]. Apart from unrolling methods, Li et al. [28] proposed a network called boundary-enhanced dual-stream network (BEDSN), which uses subnetworks to learn boundary priors and applies them in the decoding process. Nonetheless, these methods [28], [29], [30], [31] often lack the same level of mathematical interpretability provided by variational models.

In this study, we aim to develop a mathematically interpretable segmentation model that preserves MorSPs. Unlike existing methods that typically incorporate skeleton and other structural information within the loss function, such as the cldice in [10], we construct an energy functional that maintains the skeleton prior as a regularization term. This function is integrated into the soft threshold dynamics (STD) segmentation model [27]. By employing operator splitting and Fenchel–Legendre duality, we transform the model into three subproblems, for which we provide both the solution form and an iterative solving algorithm. Furthermore, we unroll the variational model into a novel network module, which we refer to as the MorSP module. This module can be seamlessly integrated into any neural network framework, endowing the network with the ability to preserve learnable skeleton priors during the decoding process. In this article, we demonstrate the effectiveness of our approach by incorporating the MorSP module to SAM, showcasing improvements in segmentation accuracy and detailed preservation for building, road, and water extraction of remote sensing images.

The main contributions of this article are as follows.

1) We provide a smooth approximation of morphological operators, offering a differentiable representation of the morphological skeleton and its variation. This allows the integration of nondifferentiable morphological operations into the backpropagation process within neural networks.
2) We design a variational segmentation model based on smooth morphological skeleton and threshold dynamics (TD), which can be efficiently solved using the operator splitting method. This facilitates the incorporation of learnable MorSPs into any neural network architecture.
3) We propose a novel approach to integrate the MorSP into the SAM. This fusion method is mathematically interpretable and can automatically learn and preserve the skeleton prior during both training and prediction across various datasets. Additionally, we demonstrate its advantages in building, road, and water extraction tasks through ablation experiments.

This article is organized as follows. Section II reviews related works, including variational segmentation models, classical morphological operators, and the SAM. Section III presents the proposed segmentation model that preserves MorSPs and introduces the MorSP module, detailing its integration into SAM. Section IV evaluates the performance of MorSP module on remote sensing images. Section V presents numerical experimental results and analysis. Finally, we summarize and discuss this work.

II. Related Works

A. Variational Segmentation Methods

Mathematically, a remote sensing image is represented as a map $I{:}\Omega \to \mathbb{R}^d$ on the pixel domain Ω, where $d = 1$ means a grayscale image, and $d > 1$ means multispectral image. Therefore, segmentation tasks can be seen to divide the pixel domain Ω into many distinct parts: $\Omega = \cup_{i=1}^{N} \Omega_i$, $\Omega_i \cap \Omega_j = \emptyset, i \neq j$, and N is the total classes. Subregion Ω_i can be represented as its relax indicative function $u_i{:}\Omega \to [0, 1]$ and the segmentation condition can be written as a simplex

$$\mathbb{U} = \left\{ \mathbf{u} = (u_1, \ldots, u_N) \in [0, 1]^N : \sum_{i=1}^{N} u_i(x) = 1 \forall x \in \Omega \right\}.$$

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The Potts model [1] is one of the most classic variational segmentation models and it solves

$$\min_{\mathbf{u}\in\mathbb{U}} -\sum_{i=1}^{N}\int_{\Omega_i} o_i(x)u_i(x)\mathrm{d}x + \lambda\sum_{i=1}^{N}|\partial\Omega_i| \tag{1}$$

where $o_i(x)$ denotes the similarity of pixel x to the ith class. The second term is a spatial regularization term, in which $\lambda > 0$ is a parameter and $|\partial\Omega_i|$ denotes the length of the boundary of Ω_i. Usually, total variation (TV) [32] is applied to represent the length term in segmentation models [33], [34].

Due to the nonsmooth and high calculation cost of TV, a TD method that expresses boundaries as differences in values within neighborhoods has been proposed. The idea of TD originally stemmed from the spatially-based Markov random field [35] and was later applied to spatial regularization in fuzzy, K-means, and Gaussian mixture model-type-based image segmentation [36], [37], [38], [39]. The advantage of the TD method is that it can be efficiently solved by the Merriman–Bence–Osher (MBO) splitting scheme [40] and applied to partial differential equation (PDE) based image segmentation [41], [42]. The TD energy to approximate the boundary length can be written as

$$|\partial\Omega_i| \approx \sqrt{\frac{\pi}{\sigma}}\sum_{i=1}^{N}\int_{\Omega} u_i(x)(f * (1-u_i))(x)\mathrm{d}x \tag{2}$$

where $*$ denotes the convolution and f is a kernel such as Gaussian function $f(x) = 1/(2\pi\sigma^2)e^{-(\|x\|^2)/(2\sigma^2)}$.

Liu et al. [27] proposed an STD model based the above TD method (2) using twice Fenchel–Legendre transformation

$$\hat{\mathbf{u}} = \arg\min_{\mathbf{u}\in\mathbb{U}}\{\langle -\mathbf{o}, \mathbf{u}\rangle + \gamma\langle \mathbf{u}, \ln\mathbf{u}\rangle + \mathcal{R}(\mathbf{u})\} \tag{3}$$

where $\mathcal{R}(\mathbf{u}) = \lambda\langle \mathbf{u}, f * (1-\mathbf{u})\rangle$. Since $\mathcal{R}$ is concave when f is a symmetric positive-definite, they offer the following iterative formula to solve this problem by using the difference of convex (DC) algorithm:

$$\mathbf{u}^{t+1} = \arg\min_{\mathbf{u}\in\mathbb{U}}\left\{\langle -\mathbf{o}, \mathbf{u}\rangle + \gamma\langle \mathbf{u}, \ln\mathbf{u}\rangle + \mathcal{R}\left(\mathbf{u}^t\right) + \left\langle \mathbf{p}^t, \mathbf{u}-\mathbf{u}^t\right\rangle\right\}$$

where $\mathbf{p}^t = \lambda f * (1-2\mathbf{u}^t) \in \partial\mathcal{R}(\mathbf{u}^t)$. This method not only has lower computational complexity and higher stability than TV regularization, but also allows the addition of other variational priors like star-shape and volume priors. Its stability and computational efficiency will greatly improve the efficiency of segmenting large-scale remote sensing images. Besides, the STD provides the relationship between the commonly used classification function softmax in neural networks and the variational method for image segmentation.

B. Classic Morphological Image Processing

The operators of mathematical morphology are more effective for extracting structural information than normal convolution operations, as they are directly related to the shapes of objects. Operators of grayscale morphology can be defined using mathematical formalism [43].

Definition 1 (Grayscale Dilation and Erosion [44]): Supposed that $I{:}\Omega \to \mathbb{R}$ is an image, structuring element $\mathbb{B}$ is a subset of Ω and $(0,0) \in \mathbb{B}$. The dilation of I by $\mathbb{B}$ is denoted by $I \oplus \mathbb{B}$ and is defined as

$$(I\oplus\mathbb{B})(x,y) := \max_{(x',y')\in\mathbb{B}}\left\{I\left(x+x', y+y'\right)\right\}.$$

Similarly, the erosion of I by $\mathbb{B}$ is denoted by $I \ominus \mathbb{B}$ and is defined as

$$(I\ominus\mathbb{B})(x,y) := \min_{(x',y')\in\mathbb{B}}\left\{I\left(x+x', y+y'\right)\right\}.$$

While dilation and erosion operators individually may not play a significant role, their combination can lead to advanced and practical algorithms. The skeleton of the gray-scale image can be represented by dilation and erosion operation

$$\mathcal{S}(I) = \sum_{j=0}^{J}\mathcal{S}_j(I) \tag{4}$$

where

$$\mathcal{S}_j(I) = \left(I\ominus^j\mathbb{B}\right) - \left(\left(I\ominus^j\mathbb{B}\right)\ominus\mathbb{B}\right)\oplus\mathbb{B}.$$

J is the times of operator composition, $\mathbb{B}$ is a structuring element, and $(I\ominus^j\mathbb{B})$ means I is eroded j times with $\mathbb{B}$ continuously. The morphological skeleton method is often used in the task of segmenting mesh or elongated structures. The rightmost column in Fig. 1 shows the skeleton extraction results of buildings and roads in the remote sensing images by the above method. However, the operators defined in morphology inherently contain nondifferentiable min and max operators, and designing them directly into network architectures would result in the absence of gradients for backpropagation. Therefore, we will address this issue in Section III-A.

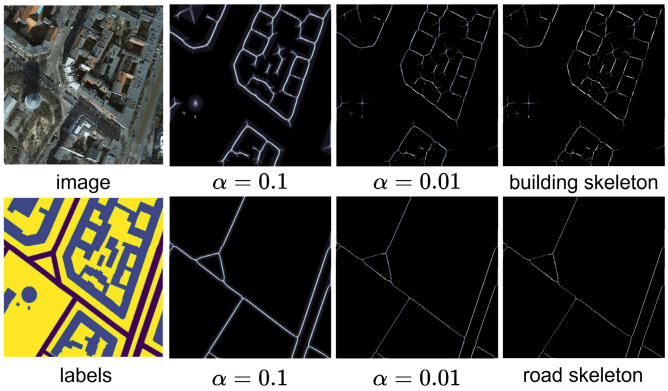


Fig. 1. Illustration of skeleton extraction: from left to right are original image and labels, smooth skeletons, and classic skeletons.

C. SAM and Fine-Tuning Strategies

SAM [15] is an interactive foundation model designed for segmentation. Trained on millions of natural images, it exhibits strong zero-shot generalization capability. The model consists of three main components: an image encoder, a prompt encoder, and a mask decoder. The image encoder is a ViT [5] pretrained using masked autoencoders (MAEs) [45], which maps input images into a high-dimensional embedding space, denoted as $\Phi_{\theta_1}{:}\mathbb{R}^{c\times h\times w} \to \mathbb{R}^{c_1\times h_1\times w_1}$. The prompt encoder converts two sets of prompts (sparse:

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points and boxes; dense: masks) to 256-D embeddings, denoted as Φ_{θ_2}:$\mathbb{R}^k \times \mathbb{R}^{h\times w} \to \mathbb{R}^{c_1\times k} \times \mathbb{R}^{c_1\times h_1\times w_1}$. The mask decoder uses a two-way transformer including cross attention and multilayer perceptron (MLP) layers to fuse image feature and prompt feature, denoted as Φ_{θ_3}:$\mathbb{R}^{c_1\times h_1\times w_1} \times \mathbb{R}^{c_1\times k} \to \mathbb{R}^{h\times w\times n}$, where n denotes the number of output. Then, we can write SAM as

$$\begin{cases} F_I = \Phi_{\theta_1}(I),\ T_p, T_m = \Phi_{\theta_2}(p, m) \\ o = \Phi_{\theta_3}\left(F_I + T_m, T_p\right) \\ u^* = \mathcal{H}(o) \end{cases} \tag{5}$$

where m and p are the dense and sparse prompt, respectively, F_I, T_m, and T_p indicate image feature, dense embedding, and sparse embedding, respectively, and o is the segmentation feature extracted by SAM. $\mathcal{H}$ is a decoding operator that is Heaviside step function here

$$\mathcal{H}(o)(x) = \begin{cases} 0, & o(x) \le 0 \\ 1, & o(x) > 0. \end{cases} \tag{6}$$

Thus, the segmentation process of SAM can be divided into three subproblems: Φ_{θ_1} and Φ_{θ_2} are the encoding processes, Φ_{θ_3} is the feature fusion process, and $\mathcal{H}$ is the final decoding process.

Pretrained visual foundational models like SAM excel at learning and extracting image features. Consequently, many studies have focused on leveraging SAM's strengths and adapting it to downstream tasks. Yan et al. [46] proposed a multimodal remote sensing segmentation model based on SAM, which can segment both optical and synthetic aperture radar (SAR) images. Chen et al. [19] developed a method to automatically segment remote sensing images by adding a prompter to generate prompts suitable for SAM.

Low-rank adaptation (LoRA) [47] is a fine-tuning strategy for large foundational models that involves freezing pretrained weights and introducing learnable low-rank decomposition matrices into each layer of transformer blocks. It offers the advantage of fewer learnable parameters while incurring no additional inference latency. For a given token $T_{\text{in}} \in \mathbb{R}^{N\times d}$ and a pretrained weight matrix $W \in \mathbb{R}^{d\times k}$, LoRA applies low-rank decomposition to update W, modifying the forward process to

$$T_{\text{out}} = T_{\text{in}}(W + \Delta W) = T_{\text{in}}(W + AB) \tag{7}$$

where $T_{\text{out}} \in \mathbb{R}^{N\times k}$ is the output token and $A \in \mathbb{R}^{d\times r}$, $B \in \mathbb{R}^{r\times k}$ are two learnable matrices with a low rank $r \ll \min(d, k)$. There have been several studies using LoRA to fine-tune SAM. For example, SAMed [48], a specialized version of SAM tailored for medical imaging, fine-tuned its image encoder using the LoRA strategy.

In this study, we mainly focus on the decoding process where the MorSP will be integrated, and the original Heaviside step function is equivalent to the minimization problem

$$\mathcal{H}(o) = \arg\min_{u\in[0,1]} \{\langle -o, u\rangle\}.$$

This equivalent optimization problem will be utilized in our subsequent method.

III. Proposed Method

A. Smooth Morphological Skeleton

In order to integrate the MorSP into our variational framework and facilitate gradient calculation, we give a smooth approximation form of the morphological operator. Below u:$\Omega \to [0, 1]$ stands for single-channel grayscale image.

Take a given closed rectangle or closed disk structuring element $\mathbb{B} \ni (0, 0)$, we note that grayscale dilation and erosion operators are

$$\mathcal{D}(u)(x) = \max_{z\in\mathbb{B}} u(x+z), \quad \mathcal{E}(u)(x) = \min_{z\in\mathbb{B}} u(x+z).$$

In order to get a smooth skeleton for gradient backpropagation in SAM, we can smooth the above operators by the log-sum-exp function.

Proposition 1 (Smooth Dilation): For $\alpha > 0$, define smooth functional

$$\mathcal{D}^\alpha(u)(x) := \alpha \ln \int_{z\in\mathbb{B}} e^{\frac{u(x+z)}{\alpha}} \mathrm{d}z \tag{8}$$

where $\alpha > 0$. If u is continuous in $x + \mathbb{B}$, we have

$$\mathcal{D}(u) = \lim_{\alpha\to 0^+} \mathcal{D}^\alpha(u).$$

Proof: The detailed proof is shown in Appendix A. ■

It is not difficulty to prove that $\mathcal{D}^\alpha(u)$ is convex with respect to u; thus, we have its twice Fenchel–Legendre transformation $(\mathcal{D}^\alpha)^{**} = \mathcal{D}^\alpha$. For a fixed point $x \in \Omega$, we define the inner product on $\mathbb{B}$

$$\langle v, u\rangle_{\mathbb{B}} := \int_{\mathbb{B}} v(x+z)u(x+z)\mathrm{d}z.$$

Proposition 2 (Smooth Dilation Operator): The twice Fenchel–Legendre transformation of $\mathcal{D}^\alpha$ is

$$(\mathcal{D}^\alpha)^{**}(u)(x) = \max_{k\in\mathbb{K}(x)} \{\langle k, u\rangle_{\mathbb{B}} - \alpha\langle k, \ln k\rangle_{\mathbb{B}}\} \quad \forall x \in \Omega$$

where $\mathbb{K}(x) = \{k$:$\Omega \to [0, 1], \int_{\mathbb{B}} k(x+z)\mathrm{d}z = 1\}$. Furthermore

$$\mathcal{D}^\alpha(u)(x) = \int_\Omega (\mathcal{K}_{\mathcal{D}}(u)(x, y)u(x+y) - \alpha\mathcal{K}_{\mathcal{D}}(u)(x, y)\ln\mathcal{K}_{\mathcal{D}}(u)(x, y))\mathrm{d}y$$

where

$$\mathcal{K}_{\mathcal{D}}(u)(x, y) = \begin{cases} \dfrac{e^{\frac{u(x+y)}{\alpha}}}{\int_{\mathbb{B}} e^{\frac{u(x+z)}{\alpha}}\mathrm{d}z}, & y \in \mathbb{B} \\ 0, & y \in \Omega\backslash\mathbb{B} \end{cases}$$

is a smooth dilation kernel.

Proof: The detailed proof is shown in Appendix B. ■

Compared with the nondifferential grayscale dilation operator in Definition 1, the above dual representation of the dilation operator is smooth, which can prevent singularities in the backpropagation of the network.

Noticing that $\mathcal{E}(u) = -\mathcal{D}(-u)$, we denote $\mathcal{E}^\alpha(u) = -\mathcal{D}^\alpha(-u)$, and we have analogous dual smooth representation

$$\mathcal{E}^\alpha(u)(x) = \int_\Omega (\mathcal{K}_{\mathcal{E}}(u)(x, y)u(x+y) + \alpha\mathcal{K}_{\mathcal{E}}(u)(x, y)\ln\mathcal{K}_{\mathcal{E}}(u)(x, y))\mathrm{d}y$$

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where

$$\mathcal{K}_{\mathcal{E}}(u)(x,y)=\begin{cases}\dfrac{e^{-\frac{u(x+y)}{\alpha}}}{\int_{\mathbb{B}} e^{-\frac{u(x+z)}{\alpha}}\mathrm{d}z}, & y\in\mathbb{B}\\ 0, & y\in\Omega\backslash\mathbb{B}\end{cases}$$

is a smooth erosion kernel.

Through duality, we unify the form of smooth morphological operators. Please note that the above $\mathcal{K}_{\mathcal{D}}$ and $\mathcal{K}_{\mathcal{E}}$ are the classic softmax operator and softmin operator, respectively; we can replace them with learnable kernel functions to implement more complex operations.

Based on the above derivation, we can use smooth operators to approximate the morphological skeleton (4) and define smooth skeleton by

$$\begin{aligned}\mathcal{S}^{\alpha}(u)&:=\mathrm{Proj}_{[0,1]}\left(\sum_{j=0}^{J}\mathcal{S}_j^{\alpha}(u)\right)\\&=\mathrm{ReLU}\left(1-\mathrm{ReLU}\left(1-\sum_{j=0}^{J}\mathcal{S}_j^{\alpha}(u)\right)\right)\\\mathcal{S}_j^{\alpha}(u)&=\underbrace{\mathcal{E}^{\alpha}\circ\mathcal{E}^{\alpha}\circ\cdots\circ\mathcal{E}^{\alpha}}_{j}(u)\\&\quad-\mathcal{D}^{\alpha}\left(\mathcal{E}^{\alpha}\left(\underbrace{\mathcal{E}^{\alpha}\circ\cdots\circ\mathcal{E}^{\alpha}}_{j}(u)\right)\right)\\&=\underbrace{\mathcal{E}^{\alpha}\circ\mathcal{E}^{\alpha}\circ\cdots\circ\mathcal{E}^{\alpha}}_{j}(u)-\mathcal{D}^{\alpha}\circ\underbrace{\mathcal{E}^{\alpha}\circ\cdots\circ\mathcal{E}^{\alpha}}_{j+1}(u).\end{aligned}$$

The above projection operator Proj is to avoid numerical overflows after smoothing. Moreover, the smooth skeleton $\mathcal{S}^{\alpha}(u)$ is differentiable.

Proposition 3: The variation of $\mathcal{S}^{\alpha}(u)$ with respect to u is

$$\begin{aligned}\frac{\delta\mathcal{S}^{\alpha}}{\delta u}&=\mathcal{H}\left(1-\mathrm{ReLU}\left(1-\sum_{j=0}^{J}\mathcal{S}_j^{\alpha}(u)\right)\right)\\&\quad\times\mathcal{H}\left(1-\sum_{j=0}^{J}\mathcal{S}_j^{\alpha}(u)\right)\sum_{j=0}^{J}\frac{\delta\mathcal{S}_j^{\alpha}}{\delta u}\end{aligned}$$

where $\mathcal{H}$ is the Heaviside step function defined in (6), and

$$\begin{aligned}\frac{\delta\mathcal{S}_j^{\alpha}}{\delta u}&=\begin{cases}\left(1-\mathcal{K}_{\mathcal{D}}\left(e^{j+1}\right)\mathcal{K}_{\mathcal{E}}\left(e^{j}\right)\right)\prod_{i=0}^{j-1}\mathcal{K}_{\mathcal{E}}\left(e^{i}\right), & j\geq 1\\ 1-\mathcal{K}_{\mathcal{D}}\left(e^{j+1}\right)\mathcal{K}_{\mathcal{E}}\left(e^{j}\right), & j=0\end{cases}\\e^{i}&=\underbrace{\mathcal{E}^{\alpha}\circ\cdots\circ\mathcal{E}^{\alpha}}_{i}(u),\quad i=1,2,\ldots,\mathcal{K}_{\mathcal{E}}\left(e^{0}\right)=\mathcal{K}_{\mathcal{E}}(u).\end{aligned}$$

Due to the smoothness of $\mathcal{E}^{\alpha}$ and $\mathcal{D}^{\alpha}$, it is easy to find that $\mathcal{S}^{\alpha}(u)$ is a smoothness of $\mathcal{S}(u)$. Fig. 1 shows the comparison of the proposed smooth skeleton with the classic morphological skeleton. Visually, the proposed skeleton is smoother, maintains better connectivity, and closely approximates the classical skeleton when α is sufficiently small.

This key formula will be used in the model that we establish in the following.

B. Segmentation Model With MorSP

Unlike methods that rely solely on loss functions such as [10], we aim to construct a variational segmentation model that enforces the segmentation results to preserve skeleton information throughout both training and prediction process. First, assume g is the ground-truth segmentation of the sample I, let us define the cost function

$$\mathcal{C}(u)=\frac{1}{2}\int_{\Omega}|\mathcal{S}^{\alpha}(u)(x)-\mathcal{S}^{\alpha}(g)(x)|^2\mathrm{d}(x)\tag{9}$$

where $\mathcal{S}^{\alpha}(u)$ and $\mathcal{S}^{\alpha}(g)$ are the skeletons of network output and ground truth g, respectively. This energy measures the difference between $\mathcal{S}^{\alpha}(u)$ and $\mathcal{S}^{\alpha}(g)$ and its variation with respect to u can be written as

$$\frac{\delta\mathcal{C}}{\delta u}=\frac{\delta\mathcal{C}}{\delta\mathcal{S}^{\alpha}}\cdot\frac{\delta\mathcal{S}^{\alpha}}{\delta u}=(\mathcal{S}^{\alpha}(u)-\mathcal{S}^{\alpha}(g))\cdot\frac{\delta\mathcal{S}^{\alpha}}{\delta u}.$$

Now, our segmentation model can be given as

$$\begin{aligned}\min_{u\in[0,1]}\{&\langle -o,u\rangle+\gamma(\langle u,\ln u\rangle+\langle 1-u,\ln(1-u)\rangle)\\&+\mathcal{R}(u)+\mathcal{C}(u)\}\end{aligned}\tag{10}$$

where $\mathcal{R}(u)=\lambda\langle u,f*(1-u)\rangle$ denotes the TD regularization term mentioned earlier in (2). By integrating the cost function (9) directly into the variational model, we ensure that essential structural details are preserved throughout the optimization process of (10).

Since the fidelity term, regularization term, and skeleton constraint term of optimization problem (10) all contain u, it is difficult to solve it directly using gradient methods. The operator splitting method [49], [50], [51] is suitable for such objective functions. Combined with augmented Lagrangian method [52], it decomposes a complex problem into two or more simple subproblems by introducing new operators. Here, we introduce a new variable w to transform the original problem into

$$\min_{u\in[0,1]}\mathcal{G}(u)+\mathcal{C}(w),\quad \text{s.t. } w-u=0$$

where $\mathcal{G}(u)=\langle -o,u\rangle+\gamma(\langle u,\ln u\rangle+\langle 1-u,\ln(1-u)\rangle)+\mathcal{R}(u)$. According to the penalty function method, we can get the object function

$$\min_{u\in[0,1],w}\mathcal{G}(u)+\mathcal{C}(w)+\eta\|w-u\|_1.$$

Here, the main reason we choose the L_1 penalty term instead of L_2 is that the dual problem of L_1 is linear with respect to u, which allows the u subproblem to have a closed-form solution.

Proposition 4 (Duality of L_1):

$$\|y\|_1=\|y\|_1^{**}=\max_{\|q\|_{\infty}\leq 1}\langle q,y\rangle.$$

Proof: The detailed proof is shown in Appendix C. ■

Now, we obtain an equivalent optimization problem

$$\min_{u\in[0,1],w}\max_{\|q\|_{\infty}\leq 1}\mathcal{G}(u)+\mathcal{C}(w)+\eta\langle q,w-u\rangle.$$

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Then, we can solve this problem by alternating optimization method to split it into the following three subproblems:

$$\begin{cases} q^{t+1} = \arg\max_{\|q\|_\infty \le 1} \{\langle q, w^t - u^t\rangle\} \\ w^{t+1} = \arg\min_w \{\mathcal{C}(w) + \eta\langle q^{t+1}, w\rangle\} \\ u^{t+1} = \arg\min_{u\in[0,1]} \{\mathcal{G}(u) - \eta\langle q^{t+1}, u\rangle\}. \end{cases}$$

We can approximately solve the first and second subproblems by projection gradient ascent and descent with one iteration and get

$$q^{t+1} \approx \text{Proj}_{\|\cdot\|_\infty \le 1}\{q^t + (w^t - u^t)\}$$
$$w^{t+1} \approx w^t - \iota\left(\frac{\delta\mathcal{C}(w)}{\delta w}\bigg|_{w^t} + \eta q^{t+1}\right).$$

For the third subproblem related to u, a difference of convex (DC) function algorithm can be employed to it by replacing concave $\mathcal{R}$ with its supporting hyperplane [27], and we get a strictly convex optimization problem

$$u^{t+1} = \arg\min_{u\in[0,1]} \left\{ \begin{array}{c} \langle -o + p^t, u\rangle + \gamma\langle u, \ln u\rangle \\ +\gamma\langle 1-u, \ln(1-u)\rangle - \eta\langle q^{t+1}, u\rangle \end{array} \right\}$$

where $p^t = \lambda f * (1 - 2u^t) \in \delta\mathcal{R}(u^t)$ is the subgradient of $\mathcal{R}(u)$. Moreover, it has a closed-form solution of sigmoid function

$$u^{t+1} = \text{sigmoid}\left(\frac{o - p^t + \eta q^{t+1}}{\gamma}\right) = \left(1 + \exp\left(-\frac{o - p^t + \eta q^{t+1}}{\gamma}\right)\right)^{-1}.$$

In summary, Algorithm 1 outlines the steps for solving the proposed model.

C. SAM With MorSP Module

Next, we will unroll Algorithm 1 into a network architecture called MorSP module that can be integrated into any segmentation neural network. Please note that in the variational model, the skeleton $\mathcal{S}^\alpha(g)$ is assumed to be known. In reality, during network prediction, such a strong prior cannot be obtained directly. Therefore, we introduce a token to learn soft skeleton label v. The architecture of the MorSP module is illustrated in Fig. 2. Specifically, this module accepts two inputs: segmentation feature o and soft skeleton label v to compute the segmentation result u, and we just take $T = 20$ layers to save computational sources.

In order to integrate MorSP module into SAM, we add a soft skeleton token to mask decoder and replace the final decoding operator $\mathcal{H}$ in SAM (5). Hence, the proposed SAM-MorSP can be written as

$$\begin{cases} F_I = \Phi_{\theta_1}(I),\ T_p, T_m = \Phi_{\theta_2}(p, m) \\ o, v = \Phi_{\theta_3}(F_I + T_m, T_p) \\ u^* = \arg\min_{u\in[0,1], w} \max_{\|q\|_\infty \le 1} \mathcal{G}(u) + \mathcal{C}(w) + \eta\langle q, w - u\rangle. \end{cases} \tag{11}$$

The third subequation in (11) represents MorSP module, which endows SAM the property of preserving skeleton prior knowledge during prediction.

Algorithm 1 Image Segmentation With MorSP

Input: segmentation feature o, skeleton label $\mathcal{S}^\alpha(g)$
Output: soft segmentation function u
Initialize:
$u^0 \leftarrow sigmoid(o)$,
$w^0 \leftarrow \frac{u^0 + \mathcal{S}^\alpha(g)}{2}$,
$q^0 \leftarrow \min(\max(w^0 - u^0, -1), 1)$.
for $t = 0, 1, 2, \ldots$
1. update dual variable q:

$$\hat{q}^{t+1} \leftarrow q^t + (w^t - u^t),$$
$$q^{t+1} \leftarrow \min(\max(\hat{q}^{t+1}, -1), 1).$$

2. update variable w:

$$w^{t+1} \leftarrow w^t - \iota\left(\frac{\delta\mathcal{C}(w)}{\delta w}\bigg|_{w^t} + \eta q^{t+1}\right).$$

3. compute the solution by DC algorithm:

$$p^t \leftarrow \lambda f * (1 - 2u^t),$$
$$u^{t+1} \leftarrow sigmoid\left(\frac{o - p^t + \eta q^{t+1}}{\gamma}\right).$$

4. Convergence check.
end

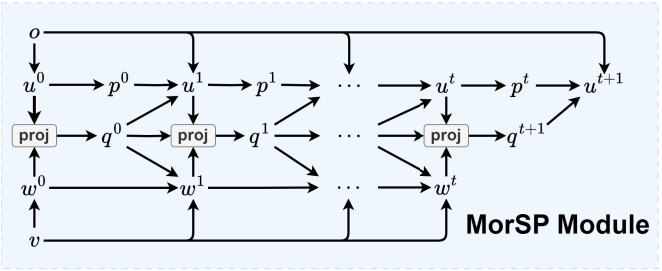


Fig. 2. Network architecture of the proposed MorSP module.

IV. Performance of MorSP Module

In this section, we design an experiment to evaluate the performance of the proposed Algorithm 1 on some remote sensing images.

We select two images of buildings and roads, and use a segmentation network to generate low-quality, rough masks for these images. We serve these rough masks as the segmentation feature o and the skeleton of ground truth as the soft skeleton $\mathcal{S}^\alpha(g)$ to postprocess the rough masks. The hyperparameters of the MorSP module are all learnable and their initial values are shown in Table I.

The results are shown in Fig. 3. Compared to original rough masks, the postprocessed outputs from the MorSP module more effectively preserve structural features, such as connectivity and holes. This confirms that our method offers significant advantages in the segmentation of remote sensing images.

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TABLE I
INITIAL HYPERPARAMETER SETTING OF MORSP MODULE

size of k	iterations	γ	λ	α	η	ι
5	20	1	1	0.05	1	1×10^{-2}

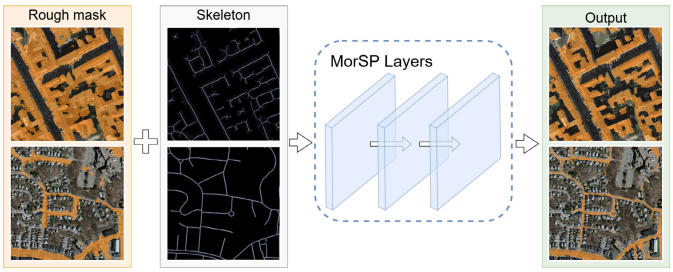


Fig. 3. Performance of the proposed MorSP module: the first column is the rough masks generated by networks, the second column is the input skeleton, and the last column is the output mask.

V. NUMERICAL EXPERIMENT I: RESULTS OF SAM-MORSP

In this section, we will train SAM-MorSP on remote sensing datasets that include buildings, roads, and water, and conduct ablation experiments to compare its performance with SAM trained using two other methods. The first method involves only fine-tuning the pretrained SAM model, and the second method employs fine-tuning with a modified skeleton loss function. The experimental results demonstrate that SAM-MorSP is superior in maintaining structural details and improving segmentation accuracy, especially in preserving the geometric integrity of remote sensing images.

A. Training Strategy and Experimental Setup

SAM provides three scales of image encoders: ViT-B, ViT-L, and ViT-H, with parameter sizes ranging from 91M to 636M. Notably, ViT-B achieves over 90% of the performance of ViT-H [15]. Thus, we select ViT-B as our encoder backbone to reduce computational demands.

To retain SAM's original generalization ability while efficiently fine-tuning it for remote sensing datasets, we freeze all image encoder's parameters and introduce LoRA layers for fine-tuning it. Then, we replace the decoder operator with the MorSP module as the final layers, so the decoder's parameters are all tunable. The architecture of the entire network is illustrated in Fig. 4.

The LoRA layers are applied to the query (Q) and value (V) components of the transformer blocks in the image encoder, with each layer consisting of two matrices of rank 4. The MorSP module's hyperparameters are all learnable and their initial values are shown in Table I. As a result, SAM-MorSP contains around 4M trainable parameters (3.9M for mask decoder, and 147k for LoRA).

The loss functions employed during training are binary cross-entropy loss

$$\mathcal{L}_{\text{CE}}(u, g) = -\langle g, \ln(u)\rangle - \langle (1-g), \ln(1-u)\rangle$$

and cl-dice loss [10]

$$\mathcal{L}_{\text{skeleton}}(v, g) = 1 - 2 \times \frac{T_{\text{prec}}(v, g) \times T_{\text{sens}}(v, g)}{T_{\text{prec}}(v, g) + T_{\text{sens}}(v, g)}$$

where

$$T_{\text{prec}} = \frac{\langle g, \mathcal{S}(v)\rangle}{\|\mathcal{S}(v)\|_1}, \quad T_{\text{sens}} = \frac{\langle \mathcal{S}(g), v\rangle}{\|\mathcal{S}(g)\|_1}.$$

We train the model with the weighted loss function

$$\mathcal{L}(u, v, g) = (1-\lambda_l)\mathcal{L}_{\text{CE}}(u, g) + \lambda_l \mathcal{L}_{\text{skeleton}}(v, g)$$

where $\lambda_l \in (0, 1)$. We utilize this loss function to fine-tune SAM with λ_l set to 0.1 during training. For comparison, we fine-tune SAM using two different methods. The first method, referred to as SAM-fine, retains the original decoder operator $\mathcal{H}$ and employs only cross-entropy loss. The second method maintains the same network as the first but utilizes a different loss function

$$\mathcal{L}(u, g) = (1-\lambda_l)\mathcal{L}_{\text{CE}}(u, g) + \lambda_l \mathcal{L}_{\text{skeleton}}(u, g)$$

and the model is denoted as SAM-cl. We train these models on a 32-GB NVIDIA Tesla V100 GPU by AdamW optimizer using the same learning strategy. Besides, the batch size is set to 1, and the initial learning rate is set to 2×10^{-4}.

B. Datasets

The proposed model is fine-tuned and evaluated on remote sensing datasets consisting of three types of objects: buildings, roads, and water, all of which contain thousands of aerial or satellite images. We apply simple preprocessing to the datasets, which is shown in Table II.

1) WHU Building Dataset [53]: The dataset was created by Wuhan University and contains manually extracted samples of buildings from aerial and satellite images. For our experiments, we selected the aerial image subset, which consists of 8189 images, each with a resolution of 512 × 512 pixels. These images were captured from Christchurch, New Zealand, and include over 220 000 individual buildings. It is widely used to train building extraction models due to its high-quality labels.

2) LAIS From Online Maps Dataset [54]: The learning aerial image segmentation (LAIS) dataset contains high-resolution aerial images of five cities from Google Maps, combined with pixel-level labels for buildings, roads, and background from OpenStreetMap (OSM). Since OSM data are generated by volunteers, these labels inevitably contain some obvious errors with an offset as large as 10 pixels. Therefore, it can be used for evaluating the performance of deep learning models in the presence of noisy data. We choose images from Berlin, Chicago, Zurich, and Potsdam, and randomly divided 30% of them into the test set for our experiment. Due to the large image size, each original image is cropped to four images and resized to 1024 × 1024 for training.

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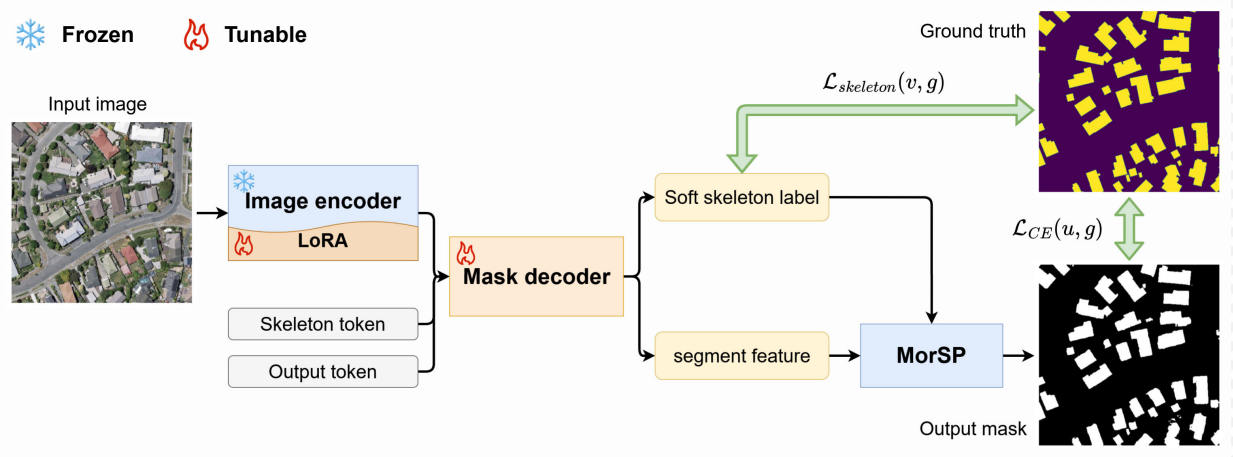


Fig. 4. Network architecture of SAM-MorSP.

TABLE II
DATA PREPROCESSING OF THE DATASETS USED

Dataset	Type	Image size	Train-Validate-Test
WHU	building	512×512	4736-1036-2416
LAIS	building & road	3000+- → 1024	2246-562-1240
Massachusetts	road	1500 → 500	2000-126-441
WHU-OPT-SAR	water	5556×3704 → 926	1800-240-360

3) Massachusetts Road Dataset [55]: The Massachusetts Road dataset consists of 1171 aerial color images with 1500 × 1500 pixels, covering urban, rural, and mountainous areas of Massachusetts, USA. The images have a relatively low resolution of 1 m, and there are some visible annotation errors in labels such as incomplete road markings. We cut each image into 500 × 500 patches and selected 2000 of them with more roads for the training set, and adopted the validate set and test set officially divided.

4) WHU-OPT-SAR Dataset [56]: This dataset contains 100 optical (OPT) images, emcompassing both RGB color and near infrared (NIR) channels, and SAR images with a resolution of 5556 × 3704 pixels in Hubei, China. In this experiment, we selected the optical images to segment the water. Each image is cropped into 24 patches with 926 × 926 pixels, and the RGB channel is input into the encoder, and the NIR channel is used as mask prompt m to input into the decoder.

C. Performances and Analyses

The performance of the above models is evaluated by four metrics including F_1-score, intersection over union (IoU), precision (Pre), and recall (Rec). We use $\Omega_u, \Omega_g \subset \Omega$ to denote the segmentation area of the network prediction and the ground truth, respectively, and the equations of evaluation metrics are as follows:

$$F_1 = 2 \times \frac{|\Omega_u \cap \Omega_g|}{|\Omega_u| + |\Omega_g|}, \quad \text{IoU} = \frac{|\Omega_u \cap \Omega_g|}{|\Omega_u \cup \Omega_g|}$$
$$\text{Pre} = \frac{|\Omega_u \cap \Omega_g|}{|\Omega_u|}, \quad \text{Rec} = \frac{|\Omega_u \cap \Omega_g|}{|\Omega_g|}$$

where $|\Omega_u|$ denotes the number of pixels contained in Ω_u. Obviously, all these metrics range from 0 to 1, with higher values indicating better segmentation performance.

The segmentation performance of the three fine-tuned models on different datasets is summarized in Tables III and IV. Notably, after fine-tuning only about 4% of the parameters of the original SAM ViT-B, all three models demonstrate strong segmentation capabilities on remote sensing datasets.

Incorporating skeleton constraint in loss function during training allows SAM-cl to achieve superior results on both the WHU and Massachusetts datasets, particularly in capturing fine targets and preserving structural details. This method effectively maintains intricate features and enhances the model's ability to delineate complex shapes with high precision. However, the cl-dice loss function faces challenges on the LAIS and WHU-OPT-SAR datasets, which contain noise and label offsets, highlighting the limited stability of such methods that rely on loss function, especially when applied to datasets with lower quality annotations.

The proposed SAM-MorSP model demonstrates superior performance across all datasets, with an approximate 2% improvement in recall compared to SAM-fine. This improvement stems from two key advantages. First, SAM-MorSP excels at detecting small buildings, which are often challenging to identify, especially in cluttered or complex environments (see Fig. 5, where the white means true positive, black means true negative, red means false positive, and green means false negative). Additionally, it performs better in segmenting objects with intricate shapes, ensuring accurate delineation of complex structures. Second, SAM-MorSP significantly enhances road and water segmentation by better preserving the connectivity (see Figs. 6 and 7), leading to more precise extraction of complex road features like intersections, roundabouts, and slender rivers. These strengths make SAM-MorSP particularly well-suited for remote sensing tasks.

D. Optimal Loss Function Weight

The weight λ_l in the loss function is a critical hyperparameter for model training, so we conduct additional experiments

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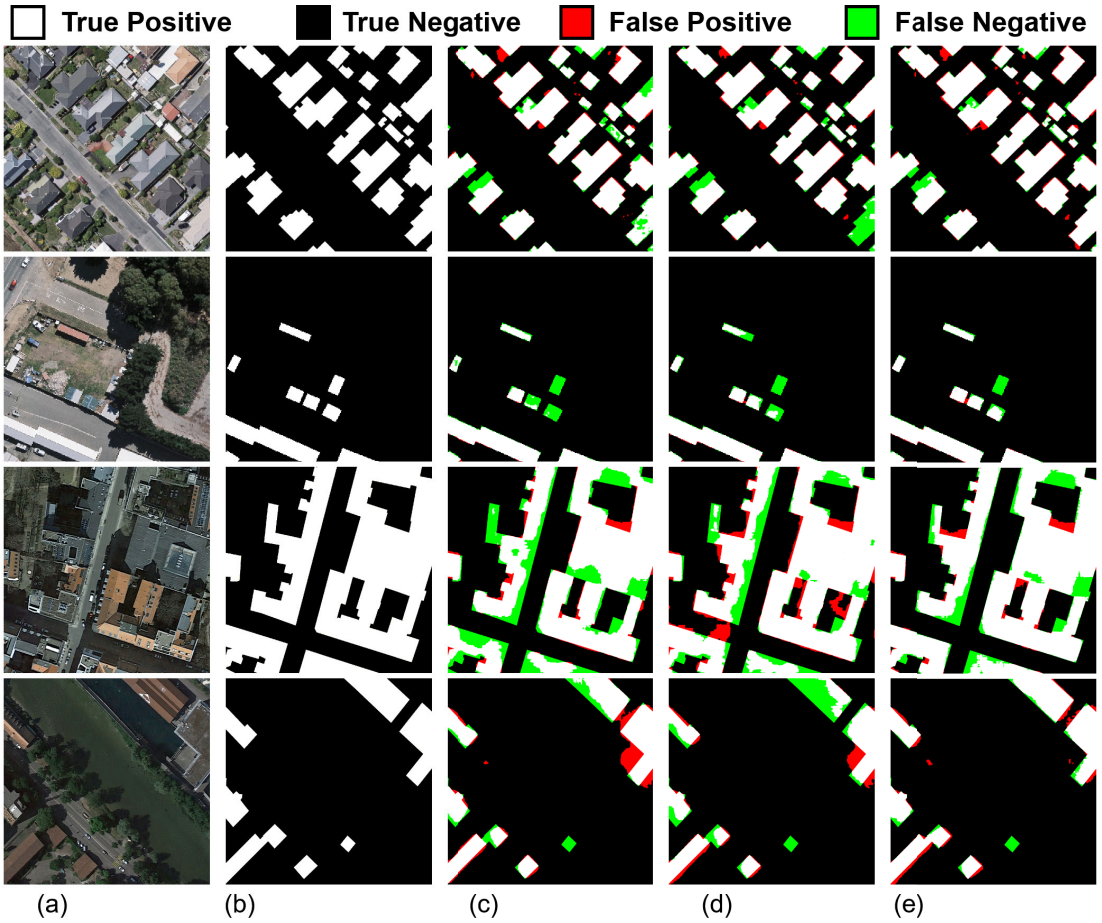


Fig. 5. Visual result of SAM-fine, SAM-cl, and SAM-MorSP on WHU and LAIS building test sets. (a) Image. (b) Ground truth. (c) SAM-fine. (d) SAM-cl. (e) SAM-MorSP.

TABLE III
COMPARISON OF SEGMENTATION METRICS ON BUILDING AND ROAD DATASETS

Type	Dataset	Model	F_1 ↑	IOU ↑	Pre ↑	Rec ↑	Dataset	Model	F_1 ↑	IOU ↑	Pre↑	Rec ↑
Building	WHU	SAM-fine	92.32	87.85	93.30	93.63	LAIS	SAM-fine	80.60	69.50	**83.39**	80.14
		SAM-cl	92.96	88.54	93.84	93.79		SAM-cl	80.54	69.16	82.09	81.23
		SAM-MorSP	**93.77**	**89.34**	**94.46**	**94.01**		SAM-MorSP	**81.39**	**70.12**	80.10	**84.70**
Road	Massachusetts	SAM-fine	74.14	60.73	**79.27**	71.53	LAIS	SAM-fine	75.33	63.35	77.09	76.62
		SAM-cl	74.67	**61.38**	79.15	72.67		SAM-cl	74.68	62.68	76.62	76.04
		SAM-MorSP	**74.98**	61.24	77.60	**74.33**		SAM-MorSP	**76.24**	**63.57**	**78.59**	**76.79**

* SAM-fine refers to the fine-tuned Segment Anything Model, and SAM-cl refers to Segment Anything Model trained with the cl-dice loss, and SAM-MorSP refers to proposed model (See Figure4). Three models have the same number of learnable parameters.

TABLE IV
COMPARISON OF SEGMENT METRICS ON WHU-OPT-SAR DATASET

Model	F_1 ↑	IOU ↑	Pre ↑	Rec ↑
SAM-fine	61.24	50.35	**79.22**	57.07
SAM-cl	59.54	47.77	65.71	**64.60**
SAM-MorSP	**62.06**	**50.94**	72.03	62.82

with two different λ_l on the Massachusetts road dataset to select the optimal weight. The results are presented in Table V, which suggests that the MorSP model consistently outperforms the baseline and as long as an appropriate weight λ_l is selected and 0.1 is likely the optimal weight.

E. Information of the Computational Demands

To fine-tune SAM, we incorporate two lightweight modules, LoRA [47] and MorSP. We evaluate the impact of these two modules on computational demands during segmenting an image with 1024 × 1024 pixels. The results are presented in Table VI.

F. Generalization Ability for Noise

To evaluate the robustness of the proposed model on noisy data, we added two levels of Gaussian noise to the test set images from the Massachusetts road dataset, with standard deviations of 0.05 and 0.1 (image grayscale range [0, 1]). The results (see Table VII) indicate that all three models experience

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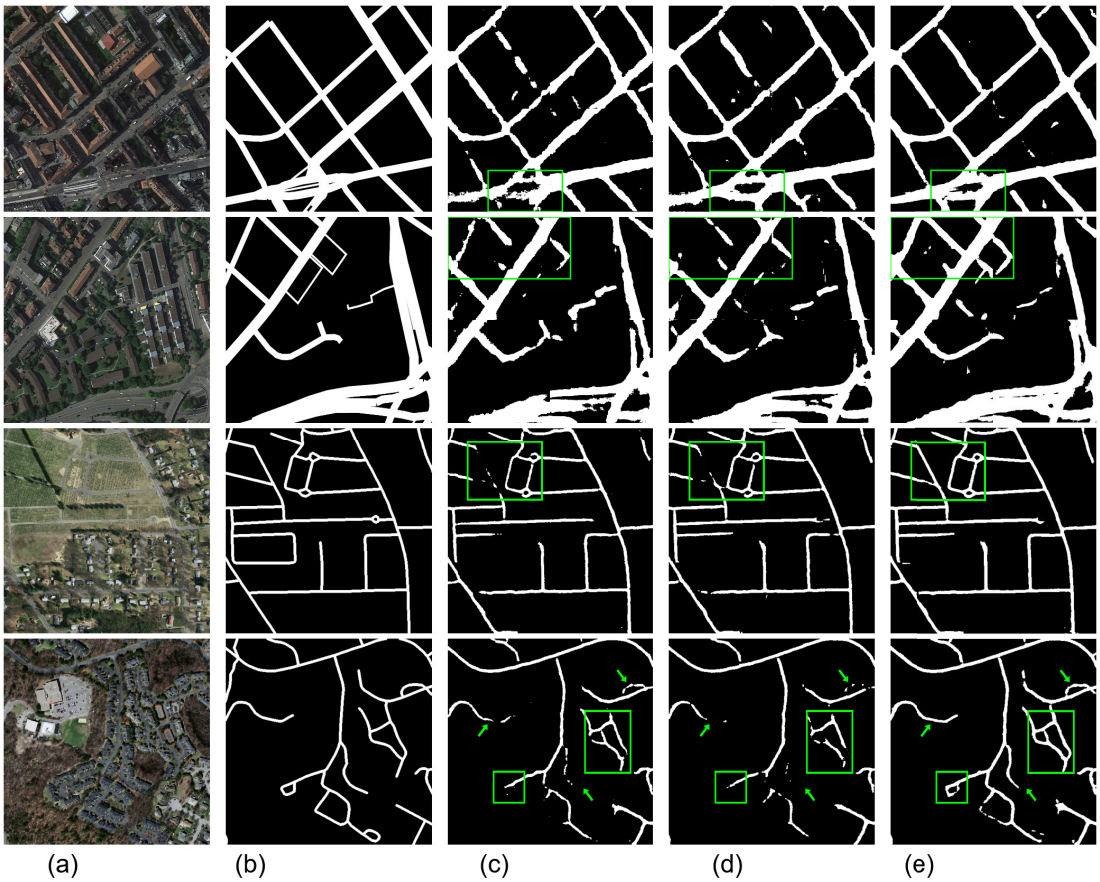


Fig. 6. Visual result of SAM-fine, SAM-cl, and SAM-MorSP on LAIS and Massachusetts road test sets. (a) Image. (b) Ground truth. (c) SAM-fine. (d) SAM-cl. (e) SAM-MorSP.

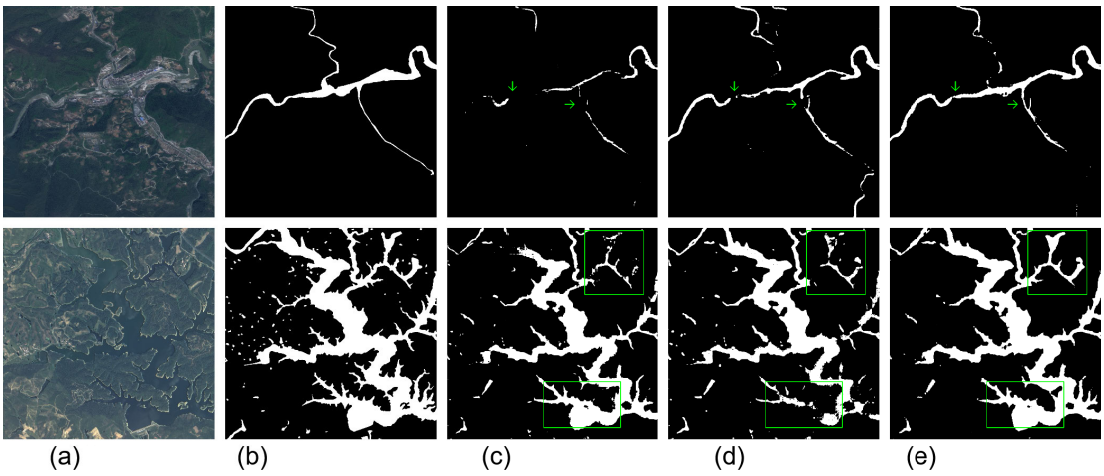


Fig. 7. Visual result of SAM-fine, SAM-cl, and SAM-MorSP on LAIS and WHU-OPT-SAR test sets. (a) Image. (b) Ground truth. (c) SAM-fine. (d) SAM-cl. (e) SAM-MorSP.

TABLE V
RESULTS OF DIFFERENT LOSS FUNCTION WEIGHTS

Weight λ_l	$F_1 \uparrow$	IOU $\uparrow$	Pre $\uparrow$	Rec $\uparrow$
0.05	74.1	**61.45**	79.87	72.56
0.1	**74.98**	61.24	77.60	**74.33**
0.2	74.13	61.44	**80.21**	72.60

TABLE VI
COMPUTATIONAL DEMANDS OF DIFFERENT MODELS

Model	Params $\downarrow$	Gflops $\downarrow$	inference time (ms) $\downarrow$
SAM (vit-b)	93.74M	976.22	153.63
+ LoRA	93.88M	977.58	156.25
+ LoRA & MorSP	93.88M	987.58	228.17

performance degradation as noise levels increase, particularly under heavy noise. The added noise obscures road features, making it challenging for the network to accurately recognize portions of the road.

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TABLE VII
TEST RESULTS ON NOISY IMAGE ON MASSACHUSETTS ROAD DATASET

Standard deviation	Model	$F_1 \uparrow$	IOU ↑	Pre ↑	Rec ↑
0.05	SAM-fine	67.34	52.30	71.97	65.89
	SAM-cl	67.46	52.63	69.45	68.67
	SAM-MorSP	**70.10**	**55.61**	**73.80**	**69.05**
0.1	SAM-fine	48.35	35.27	70.32	40.71
	SAM-cl	44.43	32.72	72.43	37.46
	SAM-MorSP	**52.30**	**40.31**	**77.04**	**44.79**

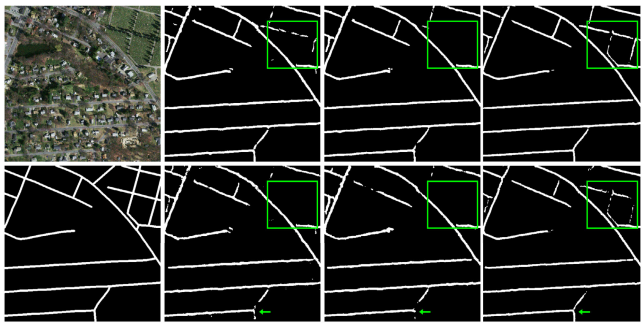


Fig. 8. Visual result of noise test. (First row, from left to right) Noisy image, mask of SAM-fine, SAM-cl, and SAM-MorSP on clean image. (Second row) Ground truth, mask of SAM-fine, SAM-cl, and SAM-MorSP on noisy image.

TABLE VIII
SEGMENTATION METRICS OF NUMERICAL EXPERIMENT ON BEDSN

Dataset	Model	$F_1 \uparrow$	IOU ↑	Pre ↑	Rec ↑
WHU	BEDSN	91.16	87.12	93.41	92.81
	+ MorSP	**92.42**	**89.00**	**94.83**	**93.58**
Massachusetts	BEDSN	79.50	59.93	**89.19**	75.73
	+MorSP	**81.20**	**61.84**	86.50	**80.21**

While SAM-cl experiences a significant decline in performance, falling below the baseline SAM-fine on heavily noisy images, the proposed SAM-MorSP model consistently outperforms both. This is attributed to the TD regularization term and smooth morphological operators, which help to smooth the image by removing small noise while preserving major structural features. Fig. 8 illustrates the comparative performance of the three methods.

VI. NUMERICAL EXPERIMENT II: MORSP ON OTHER BACKBONES

The proposed MorSP module is based on a variational model and can be integrated into any existing segmentation network. We select the BEDSN [28] as the backbone with a convolutional layer (with 1600 parameters) to learn the skeleton prior v and integrate the MorSP module at the end of the network. And the result of corresponding numerical experiments on WHU building and Massachusetts road datasets is shown in Table VIII and Fig. 9.

VII. CONCLUSION AND DISCUSSION

Objects in remote sensing images exhibit distinct geometric characteristics, featuring fine local structures and complex overall shapes. While ViT pretrained on large image datasets

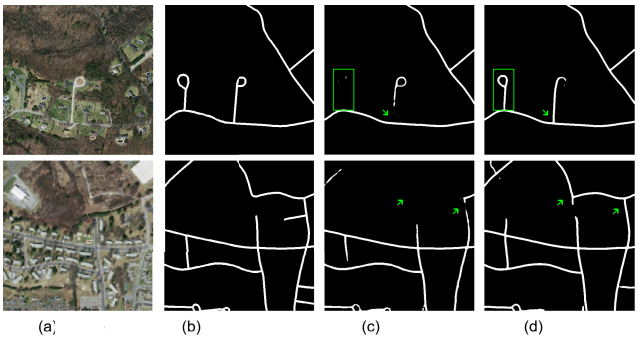


Fig. 9. Visual result of BEDSN on Massachusetts road dataset. (a) Image. (b) Ground truth. (c) BEDSN. (d) MorSP.

are powerful semantic feature extractors, they often struggle with small targets and intricate details. Recognizing that morphological skeletons can effectively describe the geometric structure and topological features of these objects, we propose a variational segmentation algorithm that integrates smooth MorSPs and unrolls it into the architecture of SAM. The novel fusion model is mathematically interpretable and constrains outputs to preserve the learned skeleton prior during both training and prediction.

The proposed method only modifies the final activation layer of the network, demonstrating superior robustness against noisy data compared to existing methods that rely on modified loss functions. This adaptability facilitates easy integration with other data-driven models, enabling the segmentation of various complex structures.

APPENDIX A
PROOF OF PROPOSITION 1

We remark that $\mathbb{M} = \{x + y : y = \arg\max_{z\in\mathbb{B}}(u)(x+z)\}$ and take $\hat{x} \in \mathbb{M}$, then

$$\mathcal{D}^{\alpha}(u)(x) = u(\hat{x}) + \alpha \ln\left(\int_{z\in\mathbb{B}} e^{\frac{u(x+z)-u(\hat{x})}{\alpha}} \mathrm{d}z\right) \leq u(\hat{x}) + \alpha \ln(|\mathbb{B}|)$$

and we have $\overline{\lim}_{\alpha\to 0^+} \mathcal{D}^{\alpha}(u)(x) \leq u(\hat{x})$. Using the definition of continuity of u at $\hat{x}$, one can prove that $\forall \epsilon > 0$, $\underline{\lim}_{\alpha\to 0^+} \mathcal{D}^{\alpha}(u)(x) \geq u(\hat{x}) - \epsilon$. From the arbitrariness of ϵ, we get $\underline{\lim}_{\alpha\to 0^+} \mathcal{D}^{\alpha}(u)(x) \geq u(\hat{x})$. Therefore

$$\overline{\lim_{\alpha\to 0^+}} \mathcal{D}^{\alpha}(u)(x) = \underline{\lim_{\alpha\to 0^+}} \mathcal{D}^{\alpha}(u)(x) = u(\hat{x}).$$

It implies that

$$\lim_{\alpha\to 0^+} \mathcal{D}^{\alpha}(u)(x) = u(\hat{x}) = \max_{z\in\mathbb{B}} u(x+z).$$

APPENDIX B
PROOF OF PROPOSITION 2

According to the definition of Fenchel–Legendre transformation, for a fixed point $x \in \Omega$

$$(\mathcal{D}^{\alpha})^{*}(k)(x) := \max_{u}\{\langle k, u\rangle_{\mathbb{B}} - \mathcal{D}^{\alpha}(u)(x)\} = \begin{cases} \alpha\langle k, \ln k\rangle_{\mathbb{B}}, & k \in \mathbb{K}(x) \\ +\infty, & \text{else} \end{cases}$$

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where $\mathbb{K}(x) = \{k{:}\Omega \to [0,1],\ \int_{\mathbb{B}} k(x+z)\mathrm{d}z = 1\},\ \forall x \in \Omega$. Therefore, we have

$$\begin{aligned}(\mathcal{D}^{\alpha})^{**}(u)(x) &:= \max_{k}\left\{\langle k, u\rangle_{\mathbb{B}} - (\mathcal{D}^{\alpha})^{*}k(x)\right\} \\ &= \max_{k\in\mathbb{K}(x)}\left\{\langle k, u\rangle_{\mathbb{B}} - \alpha\langle k, \ln k\rangle_{\mathbb{B}}\right\}\end{aligned}$$

and the above problem has a closed-form solution

$$\hat{k}(y) = \frac{e^{\frac{u(x+y)}{\alpha}}}{\int_{\mathbb{B}} e^{\frac{u(x+z)}{\alpha}}\mathrm{d}z}.$$

Appendix C
Proof of Proposition 4

We remark that $h(y) = \|y\|_1$, and its French–Legendre transform is

$$\begin{aligned}h^{*}(q) &= \max_{y}\{\langle q, y\rangle - h(y)\} \\ &= \begin{cases} +\infty, & \|q\|_{\infty} > 1 \\ 0, & \|q\|_{\infty} \le 1. \end{cases}\end{aligned}$$

Then, we have the dual norm of L_1

$$h^{**}(y) = \max_{q}\left\{\langle y, q\rangle - h^{*}(q)\right\} = \max_{\|q\|_{\infty}\le 1}\langle y, q\rangle.$$

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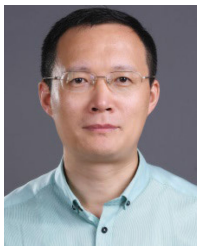
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